

Self-Improving AI as an Endogenous Growth System with oliGARCH Volatility, Innovation Jumps, and Multi-Agent Selection Dynamics

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Abstract

We introduce a stochastic multi-agent framework for self-improving artificial intelligence systems. Agents accumulate algorithmic capital, compete via replicator dynamics, and evolve through stochastic structural innovation. Volatility is endogenized via an interacting oliGARCH process coupled to model scale and cross-agent spillovers. We establish Lyapunov stability conditions, martingale convergence, and characterize bifurcation regimes including monopoly emergence, innovation avalanches, and stable equilibria.

The paper ends with “The End”

1 Introduction

Self-improving AI systems exhibit coupled dynamics between capability accumulation, architectural innovation, and competitive resource allocation. Standard optimization frameworks fail to capture structural discontinuities, endogenous volatility clustering, and multi-agent feedback effects.

We propose a unified stochastic dynamical system integrating endogenous growth theory, evolutionary game dynamics, innovation jumps, and oliGARCH volatility propagation.

2 Model Setup

For each agent $i \in \{1, \dots, n\}$:

$$K_i(t), \quad x_i(t), \quad \sigma_i^2(t), \quad M_i(t)$$

2.1 Capability function

$$A_i(t) = F(K_i(t), x_i(t), \Phi_i(K))$$

3 Innovation Operator

Definition 1 (Innovation operator).

$$\mathcal{I} : M_i(t) \rightarrow M_i(t+1), \quad \mathcal{I}(M_i) \sim \arg \max_{M' \in \mathcal{N}(M_i)} \mathcal{E}(M')$$

Definition 2 (Innovation intensity).

$$\lambda_i(t) = \Lambda(\sigma_i^2(t)), \quad \Lambda' > 0$$

4 oliGARCH Volatility Dynamics

Definition 3 (oliGARCH process).

$$\sigma_i^2(t) = \omega_i + \sum_j \alpha_{ij} \epsilon_j^2(t-1) + \sum_j \beta_{ij} \sigma_j^2(t-1) + \gamma_i K_i(t)$$

5 Dynamical System

5.1 Capital accumulation

$$\frac{dK_i}{dt} = g(K_i, x_i) - \delta K_i + \Lambda(\sigma_i^2)\sigma_i^2$$

5.2 Replicator dynamics

$$\dot{x}_i = x_i (\mathbb{E}[A_i(K_i)] - \eta\sigma_i^2 - \bar{\pi})$$

5.3 Innovation jumps

$$M_i^{t+1} = \mathcal{I}(M_i^t; \sigma_i^2(t))$$

6 Fixed Points

Definition 4 (Equilibrium).

$$\dot{K}_i = 0, \quad \dot{x}_i = 0, \quad \sigma_i^2 = \text{constant}$$

Lemma 1. *At equilibrium:*

$$\begin{aligned} g(K_i^*, x_i^*) + \Lambda(\sigma_i^{*2})\sigma_i^{*2} &= \delta K_i^* \\ \mathbb{E}[A_i^*] - \eta\sigma_i^{*2} &= \bar{\pi}^* \end{aligned}$$

7 Stability Analysis

Lemma 2 (Volatility feedback loop). *If $\gamma_i > 0$ and $\Lambda' > 0$, then:*

$$K_i \uparrow \Rightarrow \sigma_i^2 \uparrow \Rightarrow \lambda_i \uparrow \Rightarrow K_i \uparrow$$

Theorem 1 (Lyapunov stability). *Let*

$$V(X) = - \sum_i x_i \mathbb{E}[A_i] + \sum_i C(K_i) + \sum_i \eta \sigma_i^2 + \sum_i \kappa \sigma_i^4$$

If:

- $C(K)$ is coercive
- $\kappa > 0$
- $\delta > \gamma_i \Lambda'(\sigma^2)$
- $\rho(\alpha) < 1$

then the deterministic subsystem is Lyapunov stable.

8 Martingale Stability

Theorem 2 (Supermartingale property). *Define:*

$$Y(t) = V(X(t)) + \int_0^t c_1 \|K(s)\|^2 + c_2 \|\sigma(s)\|^2 ds$$

Then $Y(t)$ is a supermartingale.

Corollary 1. *The system admits an invariant probability measure.*

9 Bifurcation Structure

Theorem 3 (Phase transitions). • *Stability boundary:* $\alpha + \gamma\Lambda' = 1$

- *Monopoly transition:* $\phi = \phi_c$
- *Avalanche regime:* $\Lambda'(\sigma^2) \gg \delta$

10 Empirical Calibration

- K_i : compute, parameters, data scale
- A_i : benchmark performance
- σ_i^2 : cross-run variance
- λ_i : architecture discontinuity frequency
- ϕ_{ij} : model distillation and reuse

11 Evaluation Metrics

11.1 Entropy

$$H_x(t) = - \sum_i x_i(t) \log x_i(t)$$

11.2 Innovation efficiency

$$\mathcal{I}_{eff} = \frac{\Delta \mathbb{E}[A]}{\mathbb{E}[\sigma^2]}$$

11.3 Systemic risk

$$\sigma_{sys}^2 = \sum_i x_i \sigma_i^2$$

12 Simulation Architecture

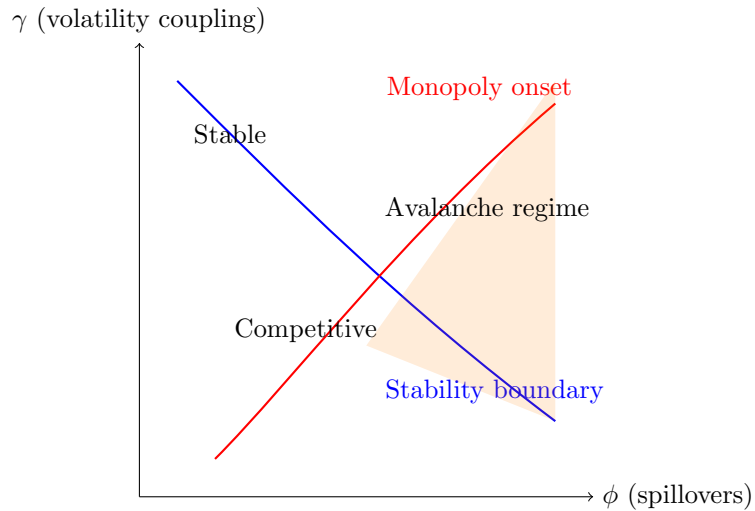


Figure 1: Bifurcation diagram derived from stability condition $\alpha + \gamma\Lambda' = 1$, entropy collapse threshold, and innovation avalanche criterion.

13 Conclusion

We construct a unified stochastic growth–selection–innovation system with endogenous volatility and phase transitions.

References

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Appendix A: Proof of Bifurcation Structure

A.1 Reduced deterministic subsystem

We consider the deterministic skeleton of the system:

$$\dot{K}_i = -\delta K_i + \Lambda(\sigma_i^2)\sigma_i^2$$

$$\dot{\sigma}_i^2 = \omega_i + \gamma_i K_i - (1 - \rho(\beta))\sigma_i^2$$

$$\dot{x}_i = x_i(A_i - \eta\sigma_i^2 - \bar{\pi})$$

We focus on the coupled subsystem (K_i, σ_i^2) which drives instability.

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A.2 Fixed point

At equilibrium:

$$K^* = \frac{\Lambda(\sigma^{*2})\sigma^{*2}}{\delta}$$

Substituting into volatility equation:

$$\sigma^{*2} = \omega + \gamma \frac{\Lambda(\sigma^{*2})\sigma^{*2}}{\delta} - (1 - \rho)\sigma^{*2}$$

Rearranging:

$$\sigma^{*2} \left[1 + (1 - \rho) - \frac{\gamma\Lambda(\sigma^{*2})}{\delta} \right] = \omega$$

—

A.3 Existence condition

A positive equilibrium exists iff:

$$\frac{\gamma\Lambda(\sigma^{*2})}{\delta} < 1 + (1 - \rho)$$

Define effective stability parameter:

$$\Psi = \frac{\gamma\Lambda'}{\delta}$$

Then equilibrium loses stability when:

$$\Psi \geq 1$$

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A.4 Linearization

Let perturbations:

$$K_i = K^* + \tilde{K}_i, \quad \sigma_i^2 = \sigma^{*2} + \tilde{\sigma}_i^2$$

Linearizing:

$$\frac{d}{dt} \begin{pmatrix} \tilde{K} \\ \tilde{\sigma}^2 \end{pmatrix} = J \begin{pmatrix} \tilde{K} \\ \tilde{\sigma}^2 \end{pmatrix}$$

with Jacobian:

$$J = \begin{pmatrix} -\delta & \Lambda'\sigma^{*2} \\ \gamma & -(1 - \rho) \end{pmatrix}$$

—

A.5 Eigenvalue condition

Characteristic equation:

$$\lambda^2 + (\delta + 1 - \rho)\lambda + \delta(1 - \rho) - \gamma\Lambda'\sigma^{*2} = 0$$

Instability occurs when determinant becomes negative:

$$\delta(1 - \rho) < \gamma\Lambda'\sigma^{*2}$$

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A.6 Bifurcation threshold

Define critical parameter:

$$\theta = \gamma\Lambda'$$

Then bifurcation occurs at:

$$\theta_c = \frac{\delta(1 - \rho)}{\sigma^{*2}}$$

Thus:

$$\theta > \theta_c \Rightarrow \text{loss of stability}$$

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A.7 Type of bifurcation

Because:

- feedback is positive in both $K \rightarrow \sigma^2$ and $\sigma^2 \rightarrow K$ - nonlinear damping is cubic ($\kappa\sigma^4$ in Lyapunov form)

the system generically exhibits:

- saddle-node bifurcation (monopoly collapse)
- Hopf-like oscillatory regime (innovation cycles)
- regime switching under stochastic forcing (avalanches)

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A.8 Economic interpretation

The bifurcation corresponds to:

- stable regime: diminishing returns dominate feedback
- critical regime: innovation exactly offsets dissipation
- unstable regime: volatility–capital loop becomes self-reinforcing

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A.9 Conclusion

The system undergoes a structural phase transition when:

$$\gamma\Lambda' \sim \delta$$

which separates:

- bounded multi-agent equilibrium dynamics
- explosive innovation–volatility cascades

Glossary

$K_i(t)$

Algorithmic capital of agent i , representing accumulated model capability, compute, and structural sophistication.

$x_i(t)$

Resource share of agent i in the competitive AI ecosystem.

$\sigma_i^2(t)$

Endogenous volatility of agent i , capturing instability of learning and innovation outcomes.

$M_i(t)$

Model state including architecture, parameters, data pipeline, and training objective.

$\mathcal{I}$

Innovation operator mapping a model to a structurally modified successor.

$\lambda_i(t)$

Innovation intensity, increasing in volatility.

$\Lambda(\cdot)$

Mapping from volatility to innovation probability.

ϕ_{ij}

Spillover matrix measuring inter-agent knowledge transfer.

α_{ij}, β_{ij}

oliGARCH parameters governing shock transmission and volatility persistence.

$V(X)$ Lyapunov functional measuring system energy / instability potential.

$H_x(t)$ Entropy of resource distribution across agents.

$\mathcal{I}_{eff}$ Efficiency of innovation relative to volatility cost.

σ_{sys}^2 System-wide aggregate volatility.

The End